Daniel Panizo

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Summary

PhD Theoretical Physicist (Uppsala University) | 7+ years research in **mathematical modeling** & analytical problem-solving | **Python** (NumPy/SciPy/Pandas, OOP), Machine Learning, Mathematica, Matlab | 5 peer-reviewed publications, $200K+ competitive funding | Rigorous **quantitative analysis** applied to challenges in technology and finance.

Technical Skills & Competencies

- **Programming:** Python (NumPy/Scipy/Pandas, OOP, scientific libraries) and ML (Scikit-learn, XGBoost). Mathematica (symbolic computations, automation, visualisation), MATLAB and LaTeX.
- **Mathematics:** Linear Algebra, advanced mathematical modeling, PDEs, non-linear and stochastic systems. High-dimensional analysis and differential geometry.
- **Methodologies:** Data analysis and visualisation, numerical simulations, algorithm development and cross-validation. Technical documentation, parameter estimation and model fitting.
- **Analytical profile:** Critical thinking, abstract reasoning, mathematical rigor, complex models synthesis and pattern identification. Scenario-based analysis and model validation. Attention to detail.
- **Languages:** *Spanish* (Native), *English* (C2), *Swedish* (B2/C1), *French* (A2) and *Japanese* (N5).

Selected Projects

- **Predictive Cosmological Model with Testable Outputs**:
  - Construction of **parameter estimation framework** linking abstract model to physical observables.
  - Delivered **quantitative predictions** testable in near-future experiments.
- **High-Dimensional Wave Perturbation Analysis**:
  - Development of **dimensional reduction algorithms** to efficiently analyse complex wave propagation in curved higher-dimensional systems.
  - **Model validation** via geometrical conservation constraints.
  - Solved complex **coupled PDE systems** with multiple perturbation modes.
- **Beanim, Python Framework for Technical Presentations**:
  - Delivered **open-source** tool for mathematical visualisation, enabling easy creation of professional slides with embedded mathematical animations.
  - Created workflow for visualising complex theoretical concepts as interactive, reusable animations.

Professional Experience & Education

2024–2026 **Postdoctoral researcher**, *Kyoto University*, Kyoto, (Japan)
- Model validation against observational data.
- Secured $100K+ funding through **competitive grants** (JSPS, Kakenhi, Daiwa)
- International collaborations and conference/seminar presentations.

2019–2024 **Ph.D. Theoretical Physics**, *Uppsala Universitet*, Uppsala, (Sweden)
- Developed **unique cosmological model** with feasible predictive power.
- **5 peer-reviewed publications** in high-impact journals.
- **Algorithm development** for model validation via high-dimensional geometrical constraints.
- **Supervision** of 3 successful master/bachelor theses. Teacher Assistant for three courses.

2016–2019 **MSc Theoretical Physics**, *Uppsala Universitet*, Uppsala, (Sweden)

2010–2016 **BSc Physics**, *Universidad Complutense de Madrid*, Madrid, (Spain)